# 02_forward_inhibition_analysis

June 15, 2026

## 1 Inhibition

Inhibition is one of biology's fundamental "brake" mechanisms: without it, biochemical and signalling networks would tend to drift towards saturation or runaway activity. But inhibition is rarely *just* a brake. When it is **nonlinear** — when the strength of inhibition depends steeply on the concentration of the inhibiting molecule — it can produce behaviours that a simple linear brake never could: sharp thresholds, switches between distinct stable states (**bistability**, or more generally **multistability**), and self-sustained **oscillations**.

This notebook explores the first of these, multistability, using a single enzymatic reaction as a worked example. The companion notebook *03_enzyme_oscillations* starts from the same kind of nonlinear inhibition kinetics, adds a delayed negative feedback loop, and shows how the same ingredients can instead produce sustained oscillations. The two notebooks therefore share a common starting point — nonlinear, Hill-type inhibition — but follow it to two different qualitative outcomes.

### 1.1 Where nonlinear inhibition comes from: enzyme kinetics

In biochemistry, inhibition is usually introduced through Michaelis–Menten kinetics: an inhibitor (I) alters the reaction rate (v) at which a substrate (S) is converted into product.

**Competitive inhibition.** The substrate and inhibitor compete for the same active site on the enzyme (E), so the inhibitor effectively reduces the amount of free enzyme available:

$$v = \frac{V_{max}[S]}{K_m\left(1 + \frac{[I]}{K_i}\right) + [S]}$$

If enough substrate is added ($[S] \to \infty$), the inhibitor's effect is overwhelmed and $v \to V_{max}$ — only the *apparent* $K_m$ increases.

**Allosteric (cooperative) inhibition.** The inhibitor binds a different site, changing the enzyme's shape so that it stops working regardless of how much substrate is present. When several inhibitor molecules bind cooperatively, this is described by a **Hill function**:

$$v = V_{max} \cdot \frac{K_i^n}{K_i^n + [I]^n}$$

where $n$ is the Hill coefficient. Plotting this for $n = 1$ gives a smooth, hyperbolic decline; for $n = 4$ it becomes sigmoidal — a "switch-like" curve where the enzyme stays fully active until a

threshold inhibitor concentration is reached, at which point activity collapses sharply. This switch-like, nonlinear shape is the key ingredient for everything that follows in this notebook.

### 1.2 Nonlinear inhibition in gene regulatory networks

The same Hill-type nonlinearity appears when one gene product represses another. If gene A inhibits gene B, and gene B inhibits gene A (mutual inhibition), the result can be a **bistable switch**: the system settles into one of two states — (high A, low B) or (low A, high B) — and stays there. This is one of the simplest models for binary cell-fate decisions, such as commitment to one of two possible cell types during development.

### 1.3 Nonlinearity elsewhere in biology

The same basic idea — a nonlinear, threshold-like inhibitory interaction — recurs across very different biological systems: in neural circuits, mutual inhibition between excitatory and inhibitory populations can generate brain-wave oscillations or sharp transitions between resting and active states; in ecology, competing species inhibit each other's growth through shared resources, with nonlinear competition terms determining whether the populations coexist or one excludes the other. The table below summarises the pattern:

| Domain | Activator/Driver | Inhibitor | Common Mathematical Tool |
|---|---|---|---|
| **Biochemistry** | Substrate (S) | Allosteric Molecule (I) | Michaelis–Menten / Hill Equation |
| **Genetics** | Transcription Factor | Repressor Protein | Sigmoidal Production Terms |
| **Neuroscience** | Glutamate / Pyramidal Cells | GABA / Interneurons | Wilson–Cowan (Sigmoid Firing Rates) |
| **Ecology** | Resource Birth Rate (r) | Competitor Species ($\alpha_{ij}$) | Lotka–Volterra Competition Terms |

In every row, the underlying mathematics is the same: a negative term that scales with the inhibitor's concentration, passed through a nonlinear (often sigmoidal) function. The rest of this notebook works through one concrete case — forward inhibition in an enzymatic reaction — in enough detail to see exactly how that nonlinearity gives rise to multistability.

## 2 Substrate Inhibition

**Substrate inhibition** (or substrate self-regulation) is a classic example of non-monotonic, nonlinear behaviour in biochemical networks. In standard Michaelis–Menten kinetics, adding more substrate always increases the reaction velocity until it plateaus at $V_{max}$. But when a substrate can bind to *both* a catalytic site and a separate regulatory (inhibitory) site on the same enzyme, the velocity curve climbs to a peak and then **falls again** as substrate concentration increases further.

### 2.1 The two-site model

Take a kinase that uses ATP as its substrate. The enzyme has two distinct binding pockets for ATP:

1. **The catalytic site** (high affinity, low capacity): binds ATP easily even at low concentrations and drives the forward (phosphorylation) reaction.
2. **The allosteric/regulatory site** (low affinity): only fills up when ATP concentrations are high. When ATP binds here, it triggers a conformational change that shuts down the catalytic site.

Schematically:

```
      + S (Fast)                + S (Slow / At High [S])
E  <=========>  ES --------> E + P     <==========>   ESS (Inactive)
    (Catalytic)    (Catalytic)          (Inhibitory)
```

where E is the free enzyme, ES is the active complex producing product P, and ESS is the dead-end, inactive complex formed when a second substrate molecule occupies the regulatory site.

### 2.2 The Haldane equation

Applying a steady-state (or quasi-equilibrium) assumption to this scheme gives the standard equation for substrate inhibition, sometimes called the **Haldane equation**:

$$v = \frac{V_{max}[S]}{K_m + [S] + \frac{[S]^2}{K_i}}$$

where $K_m$ is the dissociation constant for the catalytic site and $K_i$ is the dissociation constant for the inhibitory site.

- **At low substrate concentration** ($[S] \ll K_i$): the $[S]^2/K_i$ term is negligible, and the equation collapses to ordinary Michaelis–Menten kinetics, $v \approx \dfrac{V_{max}[S]}{K_m + [S]}$.
- **At high substrate concentration** ($[S] \gg K_m$): the $[S]^2/K_i$ term dominates the denominator, so $v \approx \dfrac{V_{max}K_i}{[S]}$ — the rate *decreases* as $[S]$ increases.

### 2.3 A bell-shaped, non-monotonic curve

The defining feature of substrate inhibition is that $v$ vs $[S]$ is **bell-shaped** rather than a saturating hyperbola. Differentiating $v$ with respect to $[S]$ and setting $dv/d[S] = 0$ shows that the peak velocity occurs at

$$[S]_{opt} = \sqrt{K_m \cdot K_i}$$

A natural computational exercise is to find $[S]_{opt}$ and the corresponding peak velocity numerically and check this relation.

### 2.4 Why would biology do this?

It may seem strange for an enzyme to "turn itself off" when its substrate is abundant, but this is a powerful regulatory strategy:

- **Overload protection.** For ATP-sensitive kinases, if ATP rises to dangerously high levels, substrate inhibition forces a slowdown of high-energy pathways, helping maintain metabolic balance.
- **Bistability and thresholding.** Combined with a downstream removal process (e.g. a phosphatase), substrate inhibition can create a **bistable toggle**: the system can sit in either a highly active or a fully dormant state, with little in between. This is the phenomenon explored in the rest of this notebook.
- **Pacemaking.** In a metabolic loop, substrate inhibition can cause the substrate concentration to spontaneously oscillate — the topic of notebook *03_enzyme_oscillations*.

## 3 Code

### 3.1 Forward Inhibition Function with Sliders

[3]:
```
from numpy import linspace
import plotly.graph_objects as go
from ipywidgets import interact, FloatSlider, VBox
import plotly.graph_objects as go

def kinetics(S, k_max, K_m, n, m):
    """
    Hill kinetics equation; reduces to Michaelis-Menten for n=1
    S: substrate concentration
    k_max: maximum reaction rate (includes enzyme concentration)
    K_m: Michaelis constant (fixed at 1)
    n: Hill exponent
    m: inhibition exponent (n<=m)
    """
    rate = (k_max * S**n) / (K_m + S**m)
    return rate

# Fixed parameters
k_max, K_m = 2, 1
# x-axis grid: substrate concentration from 0 to 4
S_values = linspace(0, 4, 100)

# Create the interactive plot
def plot_kinetics(n, m):
    """Update plot based on slider values"""
    # Evaluate the rate equation for every S in S_values, for the current (n, m)
    rate_curve = kinetics(S_values, k_max, K_m, n, m)

    fig = go.Figure()
```

```
    fig.add_trace(go.Scatter(
        x=S_values,
        y=rate_curve,
        mode='lines',
        line=dict(color='red', width=4),
        name=f'n={n:.2f}, m={m:.2f}'
    ))

    fig.update_layout(
        title=dict(
            text=f"Forward Inhibition Kinetics<br><sub>n (Hill) = {n:.2f}, m␣
 ↪(Haldane) = {m:.2f}</sub>",
            x=0.5
        ),
        xaxis=dict(title="Substrate Concentration (S)", range=[-0.1, 4.1]),
        yaxis=dict(title="Reaction Rate", range=[-0.1, 3.5]),
        width=700,
        height=600,
        showlegend=False
    )

    fig.show()

# Create sliders with ipywidgets.
# n controls the cooperativity of activation (Hill exponent);
# m controls how strongly the enzyme is inhibited at high S (Haldane/inhibition␣
 ↪exponent).
# Try n=1, m=1 first (plain Michaelis-Menten), then increase m to see the
# rate curve turn from monotonic into the bell-shaped, non-monotonic
# "substrate inhibition" curve discussed above.
interact(
    plot_kinetics,
    n=FloatSlider(
        value=1.0,
        min=1.0,
        max=5.0,
        step=0.1,
        description='n (Hill):',
        continuous_update=False,  # Only update when slider is released
        style={'description_width': '100px'}
    ),
    m=FloatSlider(
        value=1.0,
        min=0.5,
        max=4.0,
        step=0.1,
        description='m (Haldane):',
```

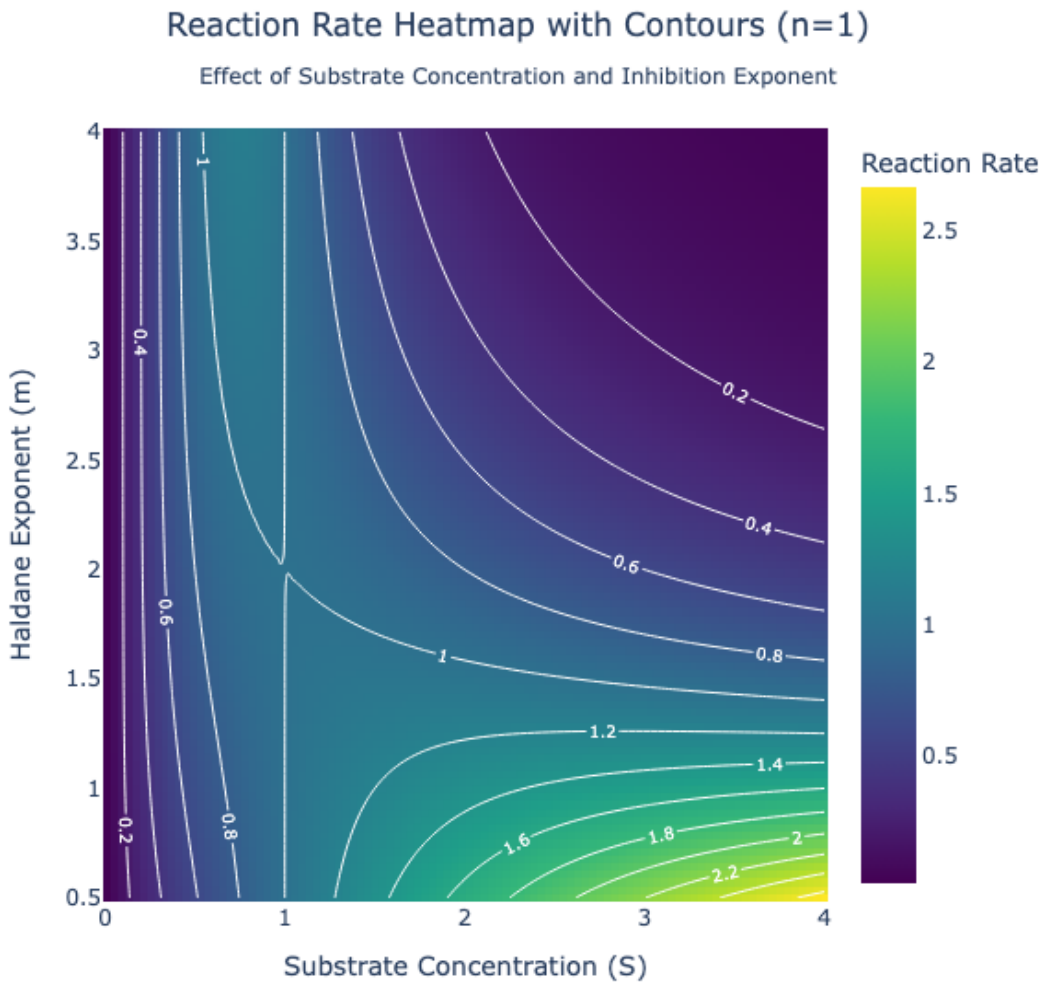
```
        continuous_update=False,
        style={'description_width': '100px'}
    )
);
```

```
interactive(children=(FloatSlider(value=1.0, continuous_update=False,␣
↪description='n (Hill):', max=5.0, min=1.…
```

### 3.2 Rate Surface

[4]:
```
from numpy import linspace, meshgrid
import plotly.graph_objects as go

def kinetics(S, k_max, K_m, n, m):
    """
    Hill kinetics equation; reduces to Michaelis-Menten for n=1
    S: substrate concentration
    k_max: maximum reaction rate (includes enzyme concentration)
    K_m: Michaelis constant (fixed at 1)
    n: Hill exponent
    m: inhibition exponent (n<=m)
    """
    rate = (k_max * S**n) / (K_m + S**m)
    return rate

# Fixed parameters: cooperativity n is fixed at 1 (plain Michaelis-Menten
# activation), so the only thing that varies below is the inhibition
# exponent m.
k_max, K_m, n = 2, 1, 1

# Create 2D grid
S_values = linspace(0.01, 4, 100)  # Start slightly above 0 to avoid division␣
↪issues
m_values = linspace(0.5, 4, 100)

# Create meshgrid: S_grid and m_grid have matching shapes so that
# kinetics() can be evaluated for every (S, m) combination at once.
S_grid, m_grid = meshgrid(S_values, m_values)

# Calculate rate for all combinations
rate_grid = kinetics(S_grid, k_max, K_m, n, m_grid)

# Combine heatmap with contour lines
fig = go.Figure()

# Add heatmap: colour encodes the reaction rate for each (S, m) pair
fig.add_trace(go.Heatmap(
    x=S_values,
```

```
    y=m_values,
    z=rate_grid,
    colorscale='Viridis',
    colorbar=dict(title="Reaction Rate"),
    hoverongaps=False,
    showscale=True
))

# Add contour lines on top, to make lines of equal rate easier to read
fig.add_trace(go.Contour(
    x=S_values,
    y=m_values,
    z=rate_grid,
    showscale=False,
    contours=dict(
        showlabels=True,
        labelfont=dict(size=9, color='white'),
        coloring='none'
    ),
    line=dict(width=1, color='white'),
    hoverinfo='skip'
))

fig.update_layout(
    title=dict(
        text="Reaction Rate Heatmap with Contours (n=1)<br><sub>Effect of␣
 ↪Substrate Concentration and Inhibition Exponent</sub>",
        x=0.5
    ),
    xaxis=dict(title="Substrate Concentration (S)"),
    yaxis=dict(title="Haldane Exponent (m)"),
    width=700,
    height=600
)

fig.show()
```

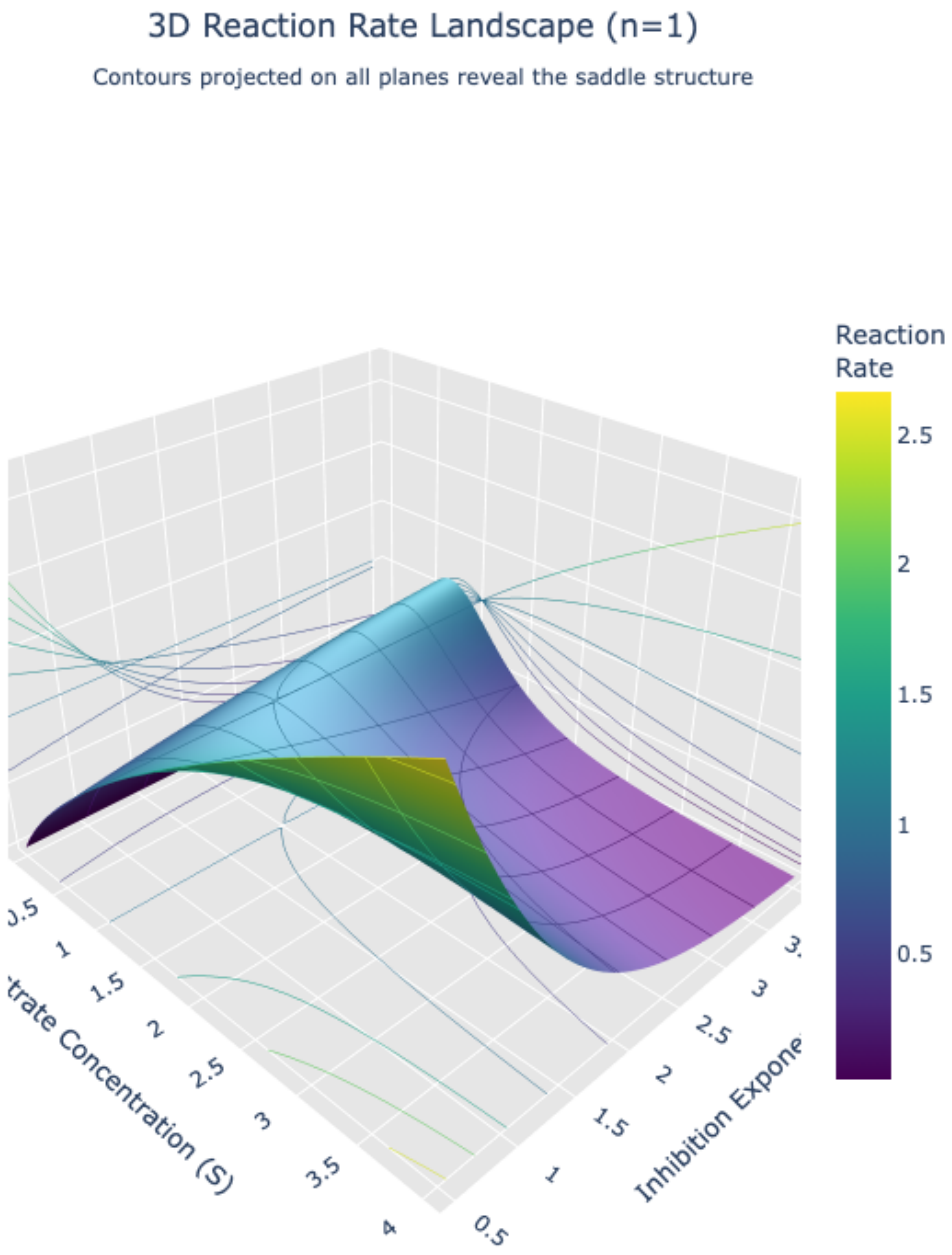



### 3.3 3D Rate Surface Plot

[5]:
```
from numpy import linspace, meshgrid
import plotly.graph_objects as go

def kinetics(S, k_max, K_m, n, m):
    """
    Hill kinetics equation; reduces to Michaelis-Menten for n=1
    S: substrate concentration
    k_max: maximum reaction rate (includes enzyme concentration)
    K_m: Michaelis constant (fixed at 1)
    n: Hill exponent
    m: inhibition exponent (n<=m)
```

```
    """
    rate = (k_max * S**n) / (K_m + S**m)
    return rate

# Fixed parameters
k_max, K_m, n = 2, 1, 1

# Create 2D grid
S_values = linspace(0.01, 4, 100)  # Start slightly above 0
m_values = linspace(0.5, 4, 100)

# Create meshgrid
S_grid, m_grid = meshgrid(S_values, m_values)

# Calculate rate for all combinations
rate_grid = kinetics(S_grid, k_max, K_m, n, m_grid)

# Create 3D surface with contours projected on all three planes.
# This is the same data as the heatmap above, but drawn as a literal
# "rate landscape": height = reaction rate, the two horizontal axes are
# substrate concentration S and inhibition exponent m.
fig = go.Figure(data=[go.Surface(
    x=S_values,
    y=m_values,
    z=rate_grid,
    colorscale='Viridis',
    colorbar=dict(title="Reaction<br>Rate", len=0.7),
    hovertemplate='S: %{x:.2f}<br>m: %{y:.2f}<br>Rate: %{z:.2f}<extra></extra>',
    lighting=dict(
        ambient=0.6,
        diffuse=0.8,
        specular=0.3,
        roughness=0.5
    ),
    contours=dict(
        x=dict(show=True, usecolormap=True, highlightcolor="white",␣
 ↪project=dict(x=True)),
        y=dict(show=True, usecolormap=True, highlightcolor="white",␣
 ↪project=dict(y=True)),
        z=dict(show=True, usecolormap=True, highlightcolor="white",␣
 ↪project=dict(z=True))
    )
)])

fig.update_layout(
    title=dict(
```

```
        text="3D Reaction Rate Landscape (n=1)<br><sub>Contours projected on␣
 ↪all planes reveal the saddle structure</sub>",
        x=0.5
    ),
    scene=dict(
        xaxis=dict(title="Substrate Concentration (S)",␣
 ↪backgroundcolor="rgb(230, 230,230)"),
        yaxis=dict(title="Inhibition Exponent (m)", backgroundcolor="rgb(230,␣
 ↪230,230)"),
        zaxis=dict(title="Reaction Rate", backgroundcolor="rgb(230, 230,230)"),
        camera=dict(
            eye=dict(x=1.5, y=-1.5, z=1.3)  # Initial viewing angle
        )
    ),
    width=900,
    height=800
)

fig.show()
```


3D Reaction Rate Landscape (n=1)
Contours projected on all planes reveal the saddle structure
Reaction Rate
2.5
2
1.5
1
0.5
trate Concentration (S)
0.5
1
1.5
2
2.5
3
3.5
4
Inhibition Expone
0.5
1
1.5
2
2.5
3

# 4 Code

## 4.1 Bistability in Open Flow

So far we have looked at how the reaction *rate* depends on substrate concentration. The key finding was that multiple binding sites — cooperativity for activation, plus a separate inhibitory site — make the rate curve strongly nonlinear and even non-monotonic (the substrate-inhibition "bell curve" from the previous section).

The next question is: what happens when this nonlinear reaction is embedded in an **open system**, with a continuous supply (influx) and removal (outflow/dilution) of substrate? Such "open flow" settings are the norm in living cells, which constantly exchange material with their environment. A real example is the ATP/PFK1 (phosphofructokinase-1) reaction in glycolysis, where ATP acts as both substrate and (via a regulatory site) inhibitor.

### 4.1.1 Why multistability matters

When the inflow/outflow balance is combined with forward inhibition, something new can happen: for the *same* parameter values, the system can have **more than one** stable steady state. This is **multistability** (with two stable states, **bistability**). Why is this biologically important?

- **Decision-making.** A bistable system can interpret a graded input (e.g. a hormone or signalling molecule concentration) as a binary output: "low" or "high". This is the basic building block of cellular decisions such as commitment to differentiation, cell-cycle progression, or apoptosis.
- **Memory.** Once a bistable system has switched to one state, it tends to *stay there* even if the input that caused the switch goes away again — a simple form of cellular memory, related to the phenomenon of **hysteresis** (the system's response depends on its history, not just its current input).
- **Robust thresholds.** Because the two stable states are separated by an unstable state acting as a threshold, small fluctuations (noise) are filtered out: the system only switches when a perturbation is large enough to cross the threshold.

### 4.1.2 Plan for this section

We will:

1. Reduce the full two-variable model to a single equation for S (decoupling it from the product P), and simulate its time course for two different values of the input parameter $a_1$ (cells below).
2. Find **all** steady states of this equation and classify each as stable or unstable, then scan over $a_1$ to build a **bifurcation diagram** — a map of how the number and position of steady states changes with $a_1$.
3. Turn the steady-state structure into a 1D **potential landscape** $V(S)$ — the "ball rolling in a landscape" picture, where stable states are valleys and unstable states are the hills between them.
4. Finally, extend this into a 2D landscape as $a_1$ varies continuously — reproducing, in a simple quantitative model, Waddington's classic "epigenetic landscape" picture of development.

[6]:
```
# solve_ivp: numerical ODE integrator, used below to simulate S(t)
from scipy.integrate import solve_ivp
# subplots: matplotlib helper for creating figure/axes grids
from matplotlib.pyplot import subplots
# linspace: evenly spaced sample points; around/var/ndarray: small numeric
# helpers used in later analysis cells
from numpy import linspace, around, var, ndarray
# find_peaks: detect local maxima in a time series (useful for spotting
# oscillations - kept here for consistency with notebook 03)
from scipy.signal import find_peaks
```

### 4.2 Feedforward inhibition in Open Flow

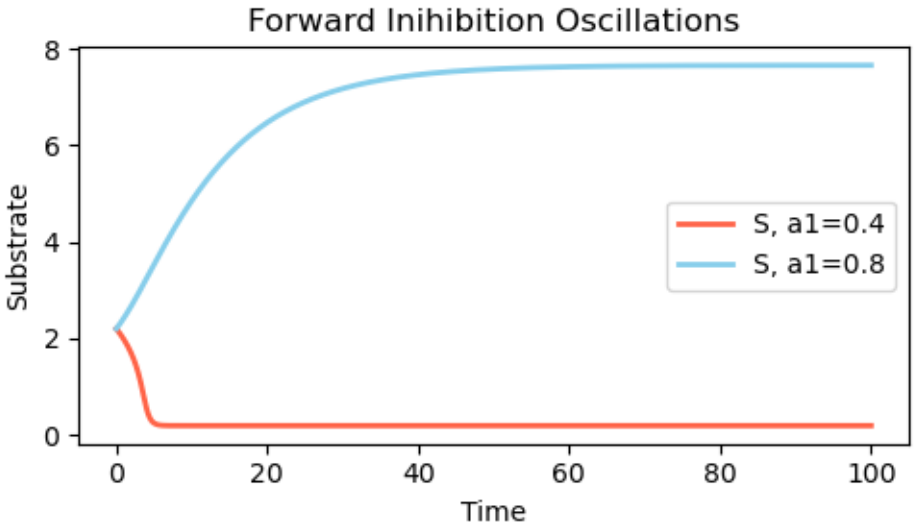
The cell below defines the reduced model: a single ODE for the substrate $S$, with constant influx, linear removal, and the same forward-inhibition kinetics used for the rate surfaces above. This is the "open flow" system referred to in the introduction to this section.

[7]:
```
def model(t, S, a1, b1, a2, b2, k_max, K_m, n, m):
    """Enzymatic Reaction with forward inhibition

    This is a 1-variable reduction of the 2-variable model used in
    notebook 03 (enzyme oscillations): here we track only the substrate S
    and ignore the downstream product P. a2 and b2 are accepted as
    arguments (so the function signature matches the 2-variable version)
    but are not used in the right-hand side below.

    t    : time (unused, but required by solve_ivp's function signature)
    S    : current substrate concentration
    a1   : constant input/production rate of S (the "control parameter"
           we vary throughout this notebook)
    b1   : linear removal rate of S (e.g. dilution, degradation)
    k_max, K_m, n, m : Hill/Haldane kinetics parameters, as in the
           kinetics() function above (forward inhibition: m > n)
    """

    # Nonlinear consumption of S by the enzyme, including forward
    # (substrate) inhibition at high S - same functional form as
    # kinetics(S, k_max, K_m, n, m) used earlier for the rate surfaces.
    enzymatic_rate = (k_max * S**n) / (K_m**m + S**m)

    # Net rate of change of S: constant inflow (a1), linear outflow
    # (b1 * S), minus the nonlinear enzymatic consumption.
    dSdt = a1 - b1 * S - enzymatic_rate

    return dSdt
```

### 4.3 Time Series

[8]:
```
# Two values of the input/production rate a1 to compare: a "low" value
# and a "high" value. We will see that they lead to qualitatively
# different long-term behaviour of S.
a1_pars  = [0.4, 0.8]

a2         = 0.01
b1, b2     = 0.1, 0.01
k_max, K_m = 2, 1
n, m       = 1, 3   # m > n: forward (substrate) inhibition is active

# Initial condition: both simulations start from the same S(0)
S_0 = 2.2

y0 = [S_0]

t_span = (0, 100)


fig, ax = subplots(figsize=(5, 3))

colors = ['tomato', 'skyblue']

# Run one simulation per value of a1 and plot S(t) for each
for index, a1 in enumerate(a1_pars):

    solution = solve_ivp(model, t_span, y0, args=(a1, b1, a2, b2, k_max, K_m,␣
 ↪n, m), method='BDF', max_step=0.1)

    t = solution.t

    S = solution.y.T

    ax.plot(t, S, label=f'S, a1={a1}', linewidth=2, color=colors[index])

ax.set_xlabel('Time')
ax.set_ylabel('Substrate')
ax.legend()
# Both curves approach a steady value (no oscillation here - that is the
# subject of notebook 03). With the same starting point S_0=2.2, the two
# values of a1 lead to very different final levels of S - a first hint of
# the bistability analysed in detail below.
ax.set_title('Forward Inhibition: Approach to Steady State')
```

```
fig.tight_layout()
```




### 4.3.1 From a single trajectory to a bifurcation diagram

The two trajectories above start from the *same* initial condition ($S_0 = 2.2$) but use different values of $a_1$, and end up at very different long-term values of $S$ (one low, one high). This already hints that the long-term behaviour depends sensitively on $a_1$ — but a single simulation per parameter value cannot tell us whether *other* steady states also exist for that same $a_1$, simply because this particular trajectory did not reach them.

To answer that question we switch from simulation to direct analysis. A steady state is any value $S^*$ for which $dS/dt = 0$; we find these by solving $f(S) = a_1 - b_1 S - (\text{enzymatic rate}) = 0$ numerically from many different starting guesses (cell below), and then classify each $S^*$ as **stable** or **unstable** by checking the sign of $df/dS$ at that point.

Repeating this for a whole range of $a_1$ values produces a **bifurcation diagram**: a plot of steady-state value $S^*$ against the control parameter $a_1$, showing how many steady states exist and how their stability changes. Wherever the *number* of steady states changes, we have crossed a **bifurcation point** — and, as we will see, this is exactly where bistability appears or disappears, and where **hysteresis** (history-dependent switching) becomes possible.

## 4.4 Analysis of Bistability

This section carries out the plan from the introduction. The code below works with the reduced, one-variable equation

$$\frac{dS}{dt} = a_1 - b_1 S - \frac{k_{max} S^n}{K_m^m + S^m}$$

(the same right-hand side as `model()` above, without the unused `a2`, `b2` arguments). For this

single equation, `find_steady_states_1D` locates every root $S^*$, `check_stability_1D` determines whether each root is stable or unstable, `compute_potential_1D` turns the equation into a potential landscape $V(S)$, and the two plotting functions visualise the results as (i) a bifurcation diagram over a range of $a_1$, and (ii) a side-by-side comparison of the phase portrait and potential for $a_1 = 0.4$ and $a_1 = 0.8$.

[9]:
```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import fsolve
from scipy.integrate import cumulative_trapezoid

def dS_dt(S, a1, b1, k_max, K_m, n, m):
    """First equation only - S dynamics independent of P"""
    enzymatic_rate = (k_max * S**n) / (K_m**m + S**m)
    return a1 - b1 * S - enzymatic_rate

def find_steady_states_1D(a1, b1, k_max, K_m, n, m):
    """Find steady states for S equation.

    A steady state S* satisfies dS/dt = 0, i.e. dS_dt(S*, ...) = 0.
    Because dS_dt is nonlinear there is no closed-form solution, so we
    search for roots numerically with fsolve, starting from many
    different initial guesses S_guess spread across the plausible range
    of S. This gives fsolve a chance to find *every* root, not just the
    one nearest to a single starting point. Converged roots are checked
    (positive, residual close to zero) and de-duplicated so that the same
    steady state is not counted twice.
    """
    steady_states = []

    # Try multiple initial guesses
    for S_guess in np.linspace(0.1, 15, 50):
        try:
            sol = fsolve(lambda S: dS_dt(S, a1, b1, k_max, K_m, n, m),
                         S_guess, full_output=True)
            if sol[2] == 1:  # Converged
                S_star = sol[0][0]
                if S_star > 0:  # Positive solution
                    residual = abs(dS_dt(S_star, a1, b1, k_max, K_m, n, m))
                    if residual < 1e-8:
                        # Check if new
                        is_new = True
                        for existing in steady_states:
                            if abs(S_star - existing) < 1e-4:
                                is_new = False
                                break
                        if is_new:
```

```
                            steady_states.append(S_star)
        except:
            pass

    return sorted(steady_states)

def check_stability_1D(S_star, a1, b1, k_max, K_m, n, m, epsilon=1e-6):
    """Check stability by examining the derivative of dS/dt at the steady state.

    Near a steady state S*, dS/dt ~ f'(S*) * (S - S*). If f'(S*) < 0, any
    small displacement decays back towards S* (stable, a "valley"); if
    f'(S*) > 0, displacements grow (unstable, a "hilltop"). f'(S*) is
    approximated here with a central finite difference.
    """
    # Compute df/dS at steady state
    f_plus = dS_dt(S_star + epsilon, a1, b1, k_max, K_m, n, m)
    f_minus = dS_dt(S_star - epsilon, a1, b1, k_max, K_m, n, m)
    derivative = (f_plus - f_minus) / (2 * epsilon)

    # Stable if derivative < 0
    is_stable = derivative < 0

    return is_stable, derivative

def compute_potential_1D(a1, b1, k_max, K_m, n, m, S_range=(0.1, 10),␣
↪resolution=1000):
    """
    Compute 1D potential V(S) such that dS/dt = -dV/dS.

    For a single ODE, dS/dt = f(S) can always be written as the negative
    gradient of a potential V(S): integrating -f(S) over S gives V(S), up
    to an arbitrary additive constant. Stable steady states (f'(S*) < 0)
    become minima ("valleys") of V, and unstable steady states become
    maxima ("hilltops"/barriers) - the "ball rolling in a landscape"
    picture used throughout this notebook.
    """
    S_array = np.linspace(S_range[0], S_range[1], resolution)

    # Compute dS/dt for all S values
    dS_dt_array = np.array([dS_dt(S, a1, b1, k_max, K_m, n, m) for S in␣
↪S_array])

    # Integrate to get potential: V(S) = - (dS/dt) dS
    potential = cumulative_trapezoid(-dS_dt_array, S_array, initial=0)

    # Shift so minimum is at zero (V is only defined up to an additive
    # constant, so this is purely for a convenient y-axis)
```

```
    potential = potential - np.min(potential)

    return S_array, potential, dS_dt_array

def plot_1D_potential_comparison():
    """Plot 1D potential for both a1 values side by side.

    Top row: phase portrait dS/dt vs S - where the curve crosses zero are
    the steady states, and the sign of dS/dt on either side shows which
    way S moves.
    Bottom row: the corresponding potential V(S) - the same information
    redrawn as a landscape (valleys = stable, hilltops = unstable).
    """

    # Parameters
    b1 = 0.1
    k_max, K_m = 2, 1
    n, m = 1, 3
    a1_values = [0.4, 0.8]

    fig, axes = plt.subplots(2, 2, figsize=(10, 7))

    colors_stable = ['red', 'blue', 'green']
    colors_unstable = ['orange']

    for idx, a1 in enumerate(a1_values):
        # Find steady states
        steady_states = find_steady_states_1D(a1, b1, k_max, K_m, n, m)

        print(f"\n{'='*50}")
        print(f"a1 = {a1}")
        print(f"{'='*50}")
        print(f"Found {len(steady_states)} steady state(s):")

        stable_states = []
        unstable_states = []

        for i, S_star in enumerate(steady_states):
            is_stable, derivative = check_stability_1D(S_star, a1, b1, k_max,␣
↪K_m, n, m)
            stability = "STABLE" if is_stable else "UNSTABLE"
            print(f"  S* = {S_star:.6f}  [{stability}]  (df/dS = {derivative:.
↪6f})")

            if is_stable:
                stable_states.append(S_star)
            else:
```

```
                unstable_states.append(S_star)



        # Compute potential and force
        S_array, potential, force = compute_potential_1D(a1, b1, k_max, K_m, n,␣
↪m)

        # Top row: Phase portrait (dS/dt vs S)
        ax_phase = axes[0, idx]
        ax_phase.plot(S_array, force, 'g-', linewidth=2.5, label='dS/dt')
        ax_phase.axhline(y=0, color='k', linestyle='--', linewidth=1, alpha=0.5)
        ax_phase.fill_between(S_array, 0, force, where=(force > 0),
                              alpha=0.3, color='blue', label='S increases')
        ax_phase.fill_between(S_array, 0, force, where=(force < 0),
                              alpha=0.3, color='red', label='S decreases')

        # Mark stable steady states
        for i, S_star in enumerate(stable_states):
            ax_phase.plot(S_star, 0, 'o', color='red', markersize=15,
                          markeredgecolor='white', markeredgewidth=2.5, zorder=5)
            # Add arrows showing stability
            ax_phase.annotate('', xy=(S_star, 0), xytext=(S_star - 0.3, 0),
                              arrowprops=dict(arrowstyle='->', color='red', lw=2))
            ax_phase.annotate('', xy=(S_star, 0), xytext=(S_star + 0.3, 0),
                              arrowprops=dict(arrowstyle='->', color='red', lw=2))

        # Mark unstable steady states
        for i, S_star in enumerate(unstable_states):
            ax_phase.plot(S_star, 0, 'X', color='cyan', markersize=15,
                          markeredgecolor='white', markeredgewidth=2.5, zorder=5)
            # Add arrows showing instability
            ax_phase.annotate('', xy=(S_star - 0.3, 0), xytext=(S_star, 0),
                              arrowprops=dict(arrowstyle='->', color='orange',␣
↪lw=2))
            ax_phase.annotate('', xy=(S_star + 0.3, 0), xytext=(S_star, 0),
                              arrowprops=dict(arrowstyle='->', color='orange',␣
↪lw=2))

        ax_phase.set_xlabel('S (Substrate)', fontsize=12, fontweight='bold')
        ax_phase.set_ylabel('dS/dt (Rate)', fontsize=12, fontweight='bold')
        ax_phase.set_title(f'Phase Portrait (a1={a1})', fontsize=13,␣
↪fontweight='bold')
        ax_phase.legend(loc='best', fontsize=9)
        ax_phase.grid(True, alpha=0.3)
```

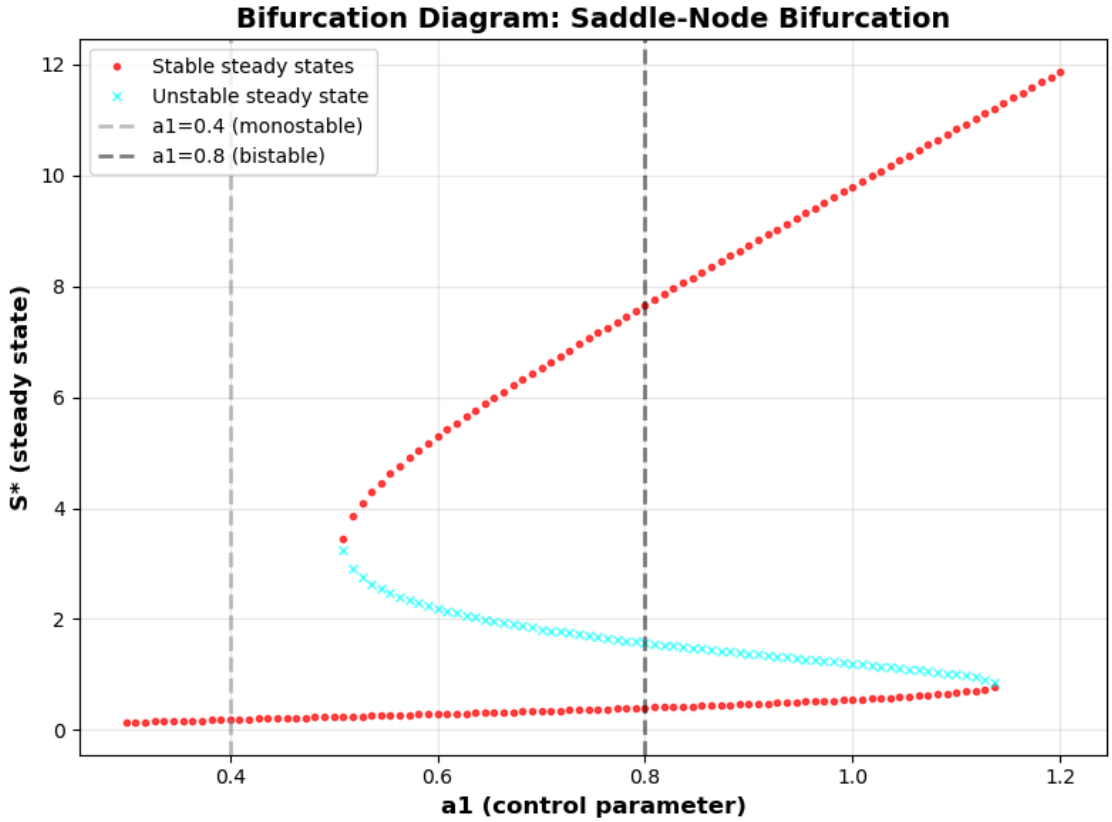
```
        # Bottom row: Potential V(S)
        ax_pot = axes[1, idx]
        ax_pot.plot(S_array, potential, 'b-', linewidth=2.5, label='V(S)')
        ax_pot.fill_between(S_array, 0, potential, alpha=0.3)

        # Mark stable steady states on potential
        for i, S_star in enumerate(stable_states):
            V_star = np.interp(S_star, S_array, potential)
            ax_pot.plot(S_star, V_star, 'o', color='red', markersize=15,
                        markeredgecolor='white', markeredgewidth=2.5,
                        label=f'Stable: S*={S_star:.3f}', zorder=5)

        # Mark unstable steady states on potential
        for i, S_star in enumerate(unstable_states):
            V_star = np.interp(S_star, S_array, potential)
            ax_pot.plot(S_star, V_star, 'X', color='cyan', markersize=15,
                        markeredgecolor='white', markeredgewidth=2.5,
                        label=f'Unstable: S*={S_star:.3f}', zorder=5)

        ax_pot.set_xlabel('S (Substrate)', fontsize=12, fontweight='bold')
        ax_pot.set_ylabel('Potential V(S)', fontsize=12, fontweight='bold')
        ax_pot.set_title(f'Potential Landscape (a1={a1})', fontsize=13,␣
 ↪fontweight='bold')
        ax_pot.legend(loc='best', fontsize=9)
        ax_pot.grid(True, alpha=0.3)
        ax_pot.set_ylim(bottom=-0.1)


    plt.tight_layout()
    # plt.savefig('potential_1D_comparison.png', dpi=300, bbox_inches='tight')
    plt.show()

def plot_combined_bifurcation():
    """Plot bifurcation diagram showing the transition from monostable to␣
 ↪bistable.

    Repeats the steady-state + stability analysis above for a whole range
    of a1 values, then plots S* against a1. Stable branches (red dots) and
    the unstable branch (cyan crosses) trace out the bifurcation diagram;
    wherever a branch appears or disappears marks a bifurcation point.
    """

    b1 = 0.1
    k_max, K_m = 2, 1
    n, m = 1, 3
```

```
# Scan a1 values
a1_values = np.linspace(0.3, 1.2, 100)

all_stable = []
all_unstable = []
all_a1_stable = []
all_a1_unstable = []

for a1 in a1_values:
    steady_states = find_steady_states_1D(a1, b1, k_max, K_m, n, m)

    for S_star in steady_states:
        is_stable, _ = check_stability_1D(S_star, a1, b1, k_max, K_m, n, m)

        if is_stable:
            all_stable.append(S_star)
            all_a1_stable.append(a1)
        else:
            all_unstable.append(S_star)
            all_a1_unstable.append(a1)

# Plot bifurcation diagram
fig, ax = plt.subplots(1, 1, figsize=(8, 6))

if all_a1_stable:
    ax.plot(all_a1_stable, all_stable, 'o', color='red',
           markersize=3, label='Stable steady states', alpha=0.7)
if all_a1_unstable and (m>2):
    ax.plot(all_a1_unstable, all_unstable, 'x', color='cyan',
           markersize=4, label='Unstable steady state', alpha=0.7)

ax.axvline(x=0.4, color='gray', linestyle='--', linewidth=2,
          alpha=0.5, label='a1=0.4 (monostable)')
if m==3:
    ax.axvline(x=0.8, color='black', linestyle='--', linewidth=2,
              alpha=0.5, label=f'a1=0.8 (bistable)')

ax.set_xlabel('a1 (control parameter)', fontsize=12, fontweight='bold')
ax.set_ylabel('S* (steady state)', fontsize=12, fontweight='bold')
ax.set_title('Bifurcation Diagram: Saddle-Node Bifurcation',
            fontsize=14, fontweight='bold')
ax.legend(loc='best', fontsize=10)
ax.grid(True, alpha=0.3)

plt.tight_layout()
# plt.savefig('bifurcation_diagram_1D.png', dpi=300, bbox_inches='tight')
plt.show()
```

```
# Run the analysis
print("="*37)
print("|| 1D POTENTIAL LANDSCAPE ANALYSIS ||")
print("="*37)
```

```
=====================================
|| 1D POTENTIAL LANDSCAPE ANALYSIS ||
=====================================
```

[10]:
```
# Generate the bifurcation diagram defined above: S* vs a1, with the
# stable branches in red and the unstable branch in cyan. See the markdown
# cell below for how to read this plot.
plot_combined_bifurcation()
```




### 4.4.1 Reading the bifurcation diagram

Running the scan over $a_1 \in [0.3, 1.2]$ shows three regimes:

- For $a_1 \lesssim 0.51$, there is a **single stable steady state** at low $S$ (e.g. at $a_1 = 0.4$, $S^* \approx 0.19$). This matches the "low" trajectory in the time-series plot above.

- Just above $a_1 \approx 0.51$, a pair of new steady states — one stable, one unstable — appears via a **saddle-node bifurcation**. From this point until $a_1 \approx 1.15$, the system has **three** steady states: a low stable state, a middle *unstable* state, and a high stable state. This is the bistable regime (e.g. at $a_1 = 0.8$: $S^* \approx 0.41$ stable, $S^* \approx 1.57$ unstable, $S^* \approx 7.66$ stable).
- Above $a_1 \approx 1.15$, the low stable branch and the unstable branch collide and annihilate in a second saddle-node bifurcation, leaving only the **high** stable steady state.

This structure is exactly what produces **hysteresis**. Imagine starting on the low branch and slowly increasing $a_1$: the system tracks the low branch smoothly until $a_1 \approx 1.15$, where that branch disappears and the system is forced to jump up to the high branch. Now slowly *decrease* $a_1$ again: the system stays on the high branch (which still exists) all the way down to $a_1 \approx 0.51$, where it disappears and the system jumps back down to the low branch. The "switch-up" and "switch-down" happen at *different* values of $a_1$ — the system's state depends on which direction it was driven from, i.e. on its history. This history-dependence is the hallmark of a bistable switch with memory, and it is the simplest dynamical mechanism underlying many biological decision-making circuits.

### 4.5 Potential and Phase Portrait

Because the reduced model is a single ODE, $dS/dt = f(S)$ can always be written as $f(S) = -\dfrac{dV}{dS}$ for some function $V(S)$ — obtained simply by integrating $-f(S)$ over $S$. This $V(S)$ is a **potential**, and it turns the abstract steady-state/stability analysis into the intuitive picture of a ball rolling on a landscape:

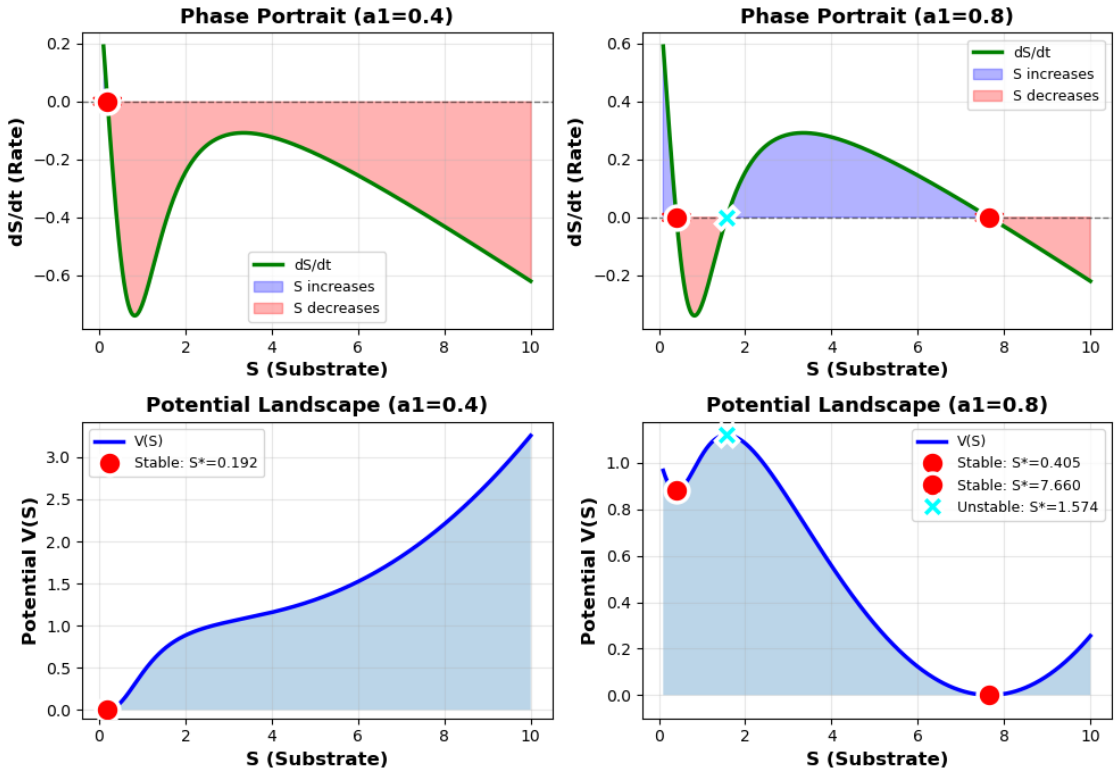
- **Stable steady states** ($f'(S^*) < 0$) are **minima** ("valleys") of $V(S)$ — a ball placed nearby rolls down into the valley and stays there.
- **Unstable steady states** ($f'(S^*) > 0$) are **maxima** ("hilltops" or barriers) of $V(S)$ — a ball placed exactly there stays only momentarily; the slightest nudge sends it rolling towards one of the neighbouring valleys.

The cell below plots, side by side, the phase portrait $dS/dt$ vs $S$ (top row) and the corresponding potential $V(S)$ (bottom row) for $a_1 = 0.4$ and $a_1 = 0.8$. This 1D landscape is the building block for the 2D "Waddington" landscape at the end of the notebook, where we let $a_1$ itself vary and watch the landscape change shape.

[11]:
```
# Generate the phase-portrait + potential-landscape comparison for
# a1 = 0.4 (monostable, single well) and a1 = 0.8 (bistable, double well).
plot_1D_potential_comparison()
```

```
================================================
a1 = 0.4
================================================
Found 1 steady state(s):
  S* = 0.191755  [STABLE]  (df/dS = -2.044283)

================================================
a1 = 0.8
================================================
Found 3 steady state(s):
```

```
S* = 0.404973   [STABLE]   (df/dS = -1.625029)
S* = 1.573804   [UNSTABLE]   (df/dS = 0.466554)
S* = 7.659890   [STABLE]   (df/dS = -0.091149)
```




### 4.5.1 Reading the potential landscape

The two panels in the bottom row make the difference between monostability and bistability visually obvious:

- $a_1 = 0.4$ **(left):** $V(S)$ has a **single well**, with its minimum at $S^* \approx 0.19$. Wherever the system starts, it rolls downhill into this one valley — there is only one possible outcome.
- $a_1 = 0.8$ **(right):** $V(S)$ has **two wells separated by a barrier**: a shallow well at $S^* \approx 0.41$ ($V \approx 0.88$), a barrier (local maximum) at $S^* \approx 1.57$ ($V \approx 1.12$), and a deeper well — the global minimum — at $S^* \approx 7.66$ ($V = 0$). Now the *initial condition* (or a transient perturbation) determines which valley the system ends up in: two different starting points, on either side of the barrier, lead to two different final states.

This is the essence of multistability as a landscape phenomenon: a smooth change in a single parameter ($a_1$, going from 0.4 to 0.8) reshapes the landscape from "one valley" to "two valleys separated by a ridge" — without any explicit switch mechanism being added to the model. The next section makes this reshaping itself the subject of the plot, by treating $a_1$ as a continuous second axis.

#### 4.5.2 From one landscape to a family of landscapes

Each panel above is a *snapshot* of the landscape for one fixed value of $a_1$. The natural next step is to ask what happens to the landscape as $a_1$ changes *continuously* — does the single well at $a_1 = 0.4$ smoothly turn into the two wells seen at $a_1 = 0.8$, and if so, how?

To find out, we compute $V(S)$ for many values of $a_1$ and stack the resulting curves side by side, turning the 1D landscape $V(S)$ into a 2D surface $V(S, a_1)$. This is exactly the construction behind Waddington's "epigenetic landscape" — one of the most influential metaphors in developmental biology.

# 5 Code

## 5.1 The Waddington Potential "Landscape"

In 1957, the developmental biologist Conrad Waddington introduced the **epigenetic landscape**: a picture of development as a ball rolling downhill through a landscape of branching valleys. Each valley represents a possible cell fate; the ridges separating valleys represent developmental decision points; and the overall shape of the landscape is sculpted by the underlying gene-regulatory network. For a long time this was a purely qualitative metaphor — there was no equation behind the picture.

The 1D potential $V(S)$ from the previous section gives us exactly the missing ingredient: an actual, computable potential derived from the kinetic equations of the system. The cell below takes the final step and promotes the control parameter $a_1$ to a second axis of the landscape, producing a genuine 2D "quasi-potential" surface $V(S, a_1)$ — a small, concrete, quantitative example of Waddington's picture.

### 5.1.1 Interpreting the axes

- $S$ (substrate concentration) is the **state** of the system — the position of the "ball".
- $a_1$ plays the role of Waddington's developmental axis or a slowly changing control signal: a morphogen concentration, an upstream regulatory input, or simply time during a developmental process. Moving along this axis reshapes the landscape itself.
- $V(S, a_1)$ is the height of the landscape: the system's state rolls downhill (towards lower $V$) at fixed $a_1$, while changes in $a_1$ can raise, lower, merge, or split the valleys.

As $a_1$ increases through $\approx 0.51$ (the saddle-node bifurcation found above), a single valley splits into two, separated by a ridge — visually, this is precisely Waddington's branch point where one developmental path divides into two.

### 5.1.2 Why this matters beyond enzyme kinetics

The same picture — valleys as stable states, ridges as decision points, and a control axis that reshapes the landscape — recurs throughout biology:

- **Gene expression and bistable switches.** In a mutual-repression "toggle switch" between two genes, the two valleys correspond to "gene A high / gene B low" and "gene A low / gene B high" expression states. A transient signal (playing the role of $a_1$ here) can push the system from one valley over the ridge into the other — after which it stays there, even once the signal is removed.

- **Development and cell-fate decisions.** In stem-cell differentiation, the valleys correspond to distinct cell types, and the changing "control parameter" corresponds to developmental signals (morphogen gradients, transcription-factor levels) that reshape the landscape over time — opening new valleys (differentiation into new cell types) or, in principle, merging them again (reprogramming/dedifferentiation).
- **Biological information processing.** A multistable landscape is, functionally, a **decision element**: a graded input is converted into a discrete output (which valley the system ends up in), and that output is *remembered* — because of hysteresis — even after the input returns to its original value. Networks built from many such elements form the basis of biological switches, memory circuits, and decision-making pathways, from cell-cycle checkpoints to immune cell activation.

The cell below builds an interactive 3D version of this landscape that can be rotated and zoomed.

[12]:
```
import numpy as np
import plotly.graph_objects as go
from scipy.optimize import fsolve
from scipy.integrate import cumulative_trapezoid

# The next four functions are the same as in the previous "Analysis of
# Bistability" section (dS_dt, find_steady_states_1D, check_stability_1D,
# compute_potential_1D) - repeated here so this cell is self-contained.
# See the markdown and comments above for the reasoning behind each one.

def dS_dt(S, a1, b1, k_max, K_m, n, m):
    """First equation only - S dynamics independent of P"""
    enzymatic_rate = (k_max * S**n) / (K_m**m + S**m)
    return a1 - b1 * S - enzymatic_rate

def find_steady_states_1D(a1, b1, k_max, K_m, n, m):
    """Find steady states for S equation (root-finding from many initial
    guesses, de-duplicated - see the detailed comments earlier in the
    notebook)."""
    steady_states = []
    for S_guess in np.linspace(0.1, 15, 50):
        try:
            sol = fsolve(lambda S: dS_dt(S, a1, b1, k_max, K_m, n, m),
                         S_guess, full_output=True)
            if sol[2] == 1:
                S_star = sol[0][0]
                if S_star > 0:
                    residual = abs(dS_dt(S_star, a1, b1, k_max, K_m, n, m))
                    if residual < 1e-8:
                        is_new = True
                        for existing in steady_states:
                            if abs(S_star - existing) < 1e-4:
                                is_new = False
                                break
```

```
                        if is_new:
                            steady_states.append(S_star)
        except:
            pass

    return sorted(steady_states)

def check_stability_1D(S_star, a1, b1, k_max, K_m, n, m, epsilon=1e-6):
    """Stable if d(dS/dt)/dS < 0 at the steady state (see earlier section)."""
    f_plus = dS_dt(S_star + epsilon, a1, b1, k_max, K_m, n, m)
    f_minus = dS_dt(S_star - epsilon, a1, b1, k_max, K_m, n, m)
    derivative = (f_plus - f_minus) / (2 * epsilon)
    is_stable = derivative < 0
    return is_stable, derivative

def compute_potential_1D(a1, b1, k_max, K_m, n, m, S_range=(0.1, 10),␣
 ↪resolution=1000):
    """Compute 1D potential V(S) such that dS/dt = -dV/dS (see earlier section).
 ↪"""
    S_array = np.linspace(S_range[0], S_range[1], resolution)
    dS_dt_array = np.array([dS_dt(S, a1, b1, k_max, K_m, n, m) for S in␣
 ↪S_array])
    potential = cumulative_trapezoid(-dS_dt_array, S_array, initial=0)
    potential = potential - np.min(potential)
    return S_array, potential, dS_dt_array

def separate_branches(S_vals, a1_vals, V_vals, stability_vals, threshold=0.5):
    """Group the (S, a1, V) steady-state points into continuous branches.

    find_steady_states_1D is called independently for each a1, so the
    resulting list of steady states is just a flat sequence of points - it
    does not yet know which points belong to the "low", "middle", or
    "high" branch, or where a branch begins/ends (at a bifurcation point).

    This function walks through the points in order and starts a new
    branch whenever either (a) the steady-state value S jumps by more than
    `threshold` from one a1 to the next, or (b) the stability flag changes.
    Both situations indicate we have crossed a bifurcation point and moved
    onto a different branch. Plotting each branch separately (rather than
    connecting all points with one line) avoids drawing spurious lines that
    jump between unrelated branches.
    """
    branches = []
    current_branch = {'S': [], 'a1': [], 'V': [], 'stable': None}

    for i in range(len(S_vals)):
        if i == 0:
```

```
            current_branch['S'].append(S_vals[i])
            current_branch['a1'].append(a1_vals[i])
            current_branch['V'].append(V_vals[i])
            current_branch['stable'] = stability_vals[i]
        else:
            # Check if this point is continuous with the current branch
            if (abs(S_vals[i] - S_vals[i-1]) < threshold and
                stability_vals[i] == current_branch['stable']):
                current_branch['S'].append(S_vals[i])
                current_branch['a1'].append(a1_vals[i])
                current_branch['V'].append(V_vals[i])
            else:
                # Start a new branch
                if len(current_branch['S']) > 0:
                    branches.append(current_branch)
                current_branch = {
                    'S': [S_vals[i]],
                    'a1': [a1_vals[i]],
                    'V': [V_vals[i]],
                    'stable': stability_vals[i]
                }

    # Don't forget the last branch
    if len(current_branch['S']) > 0:
        branches.append(current_branch)

    return branches

def create_interactive_waddington_landscape():
    """Build the 2D "Waddington" potential landscape V(S, a1) and display
    it as an interactive 3D Plotly surface.

    Overview of what happens below:
      1. For each value of a1 in a1_array, compute the 1D potential
         V(S; a1) (one row of the landscape) and find/classify its steady
         states.
      2. Stack all these rows into a 2D array `potential_matrix` -> this is
         the landscape surface.
      3. Group the steady states found across all a1 into continuous
         stable/unstable branches with separate_branches(), and draw them
         as red (stable) / cyan dashed (unstable) curves on top of the
         surface - this traces out the bifurcation diagram *on* the
         landscape.
      4. Add two reference curves showing the 1D cross-sections at
         a1=0.4 (monostable) and a1=0.8 (bistable), matching the panels
         plotted earlier.
    """
```

```
    # Parameters
    b1 = 0.1
    k_max, K_m = 2, 1
    n, m = 1, 3

    # Create arrays with high resolution
    a1_array = np.linspace(0.4, 0.8, 80)
    S_range = (0.05, 8)  # FIXED: Start from much lower value to see low steady␣
↪state
    S_resolution = 500

    # Initialize potential matrix
    S_array = np.linspace(S_range[0], S_range[1], S_resolution)
    potential_matrix = np.zeros((len(a1_array), S_resolution))

    # Store steady states
    all_steady_states = []
    all_a1_for_states = []
    all_stability = []
    all_V_for_states = []

    print("Computing interactive Waddington landscape...")
    for i, a1 in enumerate(a1_array):
        # One "row" of the landscape: the 1D potential at this a1
        _, potential, _ = compute_potential_1D(a1, b1, k_max, K_m, n, m,
                                               S_range, S_resolution)
        potential_matrix[i, :] = potential

        # Steady states (and their stability/potential value) at this a1 -
        # these will be drawn as the bifurcation curve on the landscape
        steady_states = find_steady_states_1D(a1, b1, k_max, K_m, n, m)
        for S_star in steady_states:
            is_stable, _ = check_stability_1D(S_star, a1, b1, k_max, K_m, n, m)
            V_star = np.interp(S_star, S_array, potential)
            all_steady_states.append(S_star)
            all_a1_for_states.append(a1)
            all_stability.append(is_stable)
            all_V_for_states.append(V_star)

    # Create meshgrid
    S_mesh, a1_mesh = np.meshgrid(S_array, a1_array)

    # Cap the potential for better visualization
    V_plot = potential_matrix.copy()
    V_max = np.percentile(V_plot, 98)
    V_plot = np.clip(V_plot, 0, V_max)
```

```
    # FIXED: Separate branches to avoid connecting discontinuous points
    branches = separate_branches(
        all_steady_states,
        all_a1_for_states,
        all_V_for_states,
        all_stability,
        threshold=0.5  # Adjust this if needed
    )

    # Create the figure
    fig = go.Figure()

    # Add the main surface
    fig.add_trace(go.Surface(
        x=S_mesh,
        y=a1_mesh,
        z=V_plot,
        colorscale='Viridis',
        opacity=0.95,
        name='Potential Landscape',
        colorbar=dict(
            title=dict(
                text='Potential V',
                font=dict(size=14, family='Arial, sans-serif')
            ),
            tickfont=dict(size=12),
            len=0.75,
            thickness=20
        ),
        contours=dict(
            z=dict(
                show=True,
                usecolormap=True,
                highlightcolor="white",
                project=dict(z=True)
            )
        ),
        hovertemplate='<b>S</b>: %{x:.3f}<br><b>a1</b>: %{y:.3f}<br><b>V</b>:␣
↪%{z:.3f}<extra></extra>'
    ))

    # FIXED: Add each branch separately as lines+markers
    for i, branch in enumerate(branches):
        if branch['stable']:
            fig.add_trace(go.Scatter3d(
                x=branch['S'],
                y=branch['a1'],
```

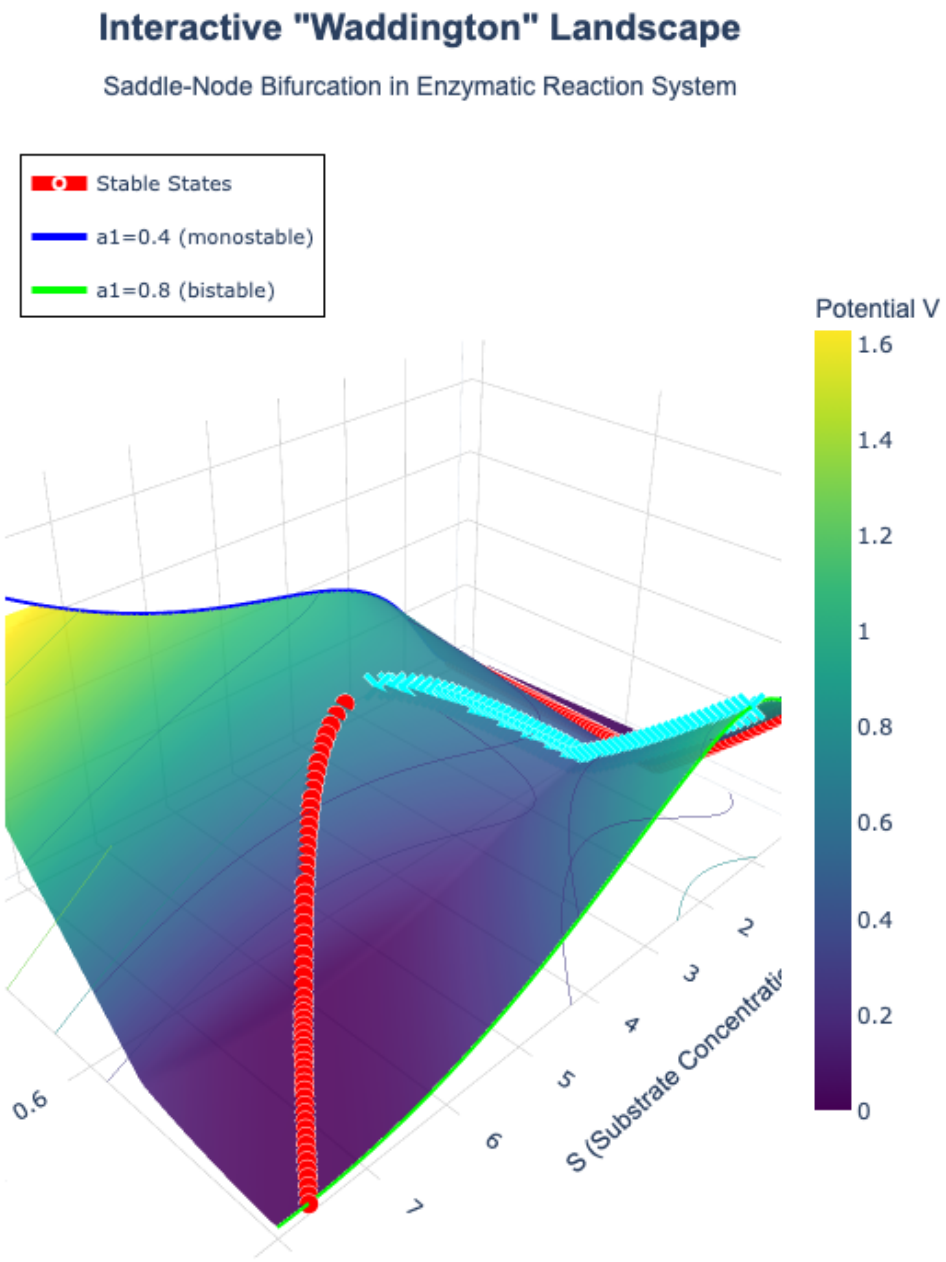
```
                z=branch['V'],
                mode='lines+markers',
                name=f'Stable Branch {i+1}' if i > 0 else 'Stable States',
                line=dict(color='red', width=8),
                marker=dict(
                    size=6,
                    color='red',
                    symbol='circle',
                    line=dict(color='white', width=2)
                ),
                showlegend=(i == 0),  # Only show legend for first stable branch
                legendgroup='stable',
                hovertemplate='<b>Stable State</b><br>S: %{x:.4f}<br>a1: %{y:.
↪4f}<br>V: %{z:.4f}<extra></extra>'
            ))
        else:
            fig.add_trace(go.Scatter3d(
                x=branch['S'],
                y=branch['a1'],
                z=branch['V'],
                mode='lines+markers',
                name=f'Unstable Branch {i+1}' if i > 0 else 'Unstable States',
                line=dict(color='cyan', width=8, dash='dash'),
                marker=dict(
                    size=4,
                    color='cyan',
                    symbol='x',
                    line=dict(color='white', width=2)
                ),
                showlegend=(i == 0),  # Only show legend for first unstable␣
↪branch
                legendgroup='unstable',
                hovertemplate='<b>Unstable State</b><br>S: %{x:.4f}<br>a1: %{y:.
↪4f}<br>V: %{z:.4f}<extra></extra>'
            ))

    # Add reference planes
    idx_04 = np.argmin(np.abs(a1_array - 0.4))
    S_plane = np.linspace(S_range[0], S_range[1], 50)
    a1_plane_04 = np.full_like(S_plane, 0.4)
    V_plane_04 = np.interp(S_plane, S_array, potential_matrix[idx_04, :])

    fig.add_trace(go.Scatter3d(
        x=S_plane,
        y=a1_plane_04,
        z=V_plane_04,
        mode='lines',
```

```
        name='a1=0.4 (monostable)',
        line=dict(color='blue', width=4),
        hovertemplate='<b>a1=0.4</b><br>S: %{x:.3f}<br>V: %{z:.3f}<extra></
↪extra>'
    ))

    idx_06 = np.argmin(np.abs(a1_array - 0.8))
    a1_plane_06 = np.full_like(S_plane, 0.8)
    V_plane_06 = np.interp(S_plane, S_array, potential_matrix[idx_06, :])

    fig.add_trace(go.Scatter3d(
        x=S_plane,
        y=a1_plane_06,
        z=V_plane_06,
        mode='lines',
        name='a1=0.8 (bistable)',
        line=dict(color='lime', width=4),
        hovertemplate='<b>a1=0.8</b><br>S: %{x:.3f}<br>V: %{z:.3f}<extra></
↪extra>'
    ))

    # Update layout
    fig.update_layout(
        title=dict(
            text='<b>Interactive "Waddington" Landscape</b><br>' +
                 '<sub>Saddle-Node Bifurcation in Enzymatic Reaction System</
↪sub>',
            x=0.5,
            xanchor='center',
            font=dict(size=20, family='Arial, sans-serif')
        ),
        scene=dict(
            xaxis=dict(
                title=dict(text='S (Substrate Concentration)',
                           font=dict(size=14, family='Arial, sans-serif')),
                gridcolor='lightgray',
                showbackground=True,
                backgroundcolor='rgba(230, 230, 230, 0.5)'
            ),
            yaxis=dict(
                title=dict(text='a1 "Time"',
                           font=dict(size=14, family='Arial, sans-serif')),
                gridcolor='lightgray',
                showbackground=True,
                backgroundcolor='rgba(230, 230, 230, 0.5)'
            ),
            zaxis=dict(
```

```
                title=dict(text='Potential V(S, a1)',
                           font=dict(size=14, family='Arial, sans-serif')),
                gridcolor='lightgray',
                showbackground=True,
                backgroundcolor='rgba(230, 230, 230, 0.5)'
            ),
            camera=dict(
                eye=dict(x=1.5, y=1.5, z=1.3),
                center=dict(x=0, y=0, z=0)
            ),
            aspectmode='manual',
            aspectratio=dict(x=1.5, y=1, z=0.8)
        ),
        width=1200,
        height=800,
        showlegend=True,
        legend=dict(
            x=0.02,
            y=0.98,
            bgcolor='rgba(255, 255, 255, 0.8)',
            bordercolor='black',
            borderwidth=1,
            font=dict(size=11)
        ),
        hovermode='closest',
        template='plotly_white'
    )

    # Save to HTML
    html_file = 'waddington_landscape_interactive.html'
    fig.write_html(html_file)
    print(f"\n  Interactive plot saved to: {html_file}")

    # Show the plot
    fig.show()

    return fig

# Run the interactive Plotly analysis
print("="*60)
print("INTERACTIVE 'WADDINGTON' LANDSCAPE")
print("="*60)
fig1 = create_interactive_waddington_landscape()
print("\n" + "="*60)
print("EXPORT OPTIONS:")
print("="*60)
print("1. Open 'waddington_landscape_interactive.html' in your browser")
```

```
print("2. Interactive controls:")
print("   - Rotate: Click and drag")
print("   - Zoom: Scroll or pinch")
print("   - Pan: Right-click and drag")
print("   - Reset view: Double-click")
print("3. Download static PNG using camera icon in toolbar")
print("4. HTML file is self-contained and shareable!")
print("="*60)
```

```
============================================================
INTERACTIVE 'WADDINGTON' LANDSCAPE
============================================================
Computing interactive Waddington landscape…

  Interactive plot saved to: waddington_landscape_interactive.html
```




==========

EXPORT OPTIONS:

==========

```
1. Open 'waddington_landscape_interactive.html' in your browser
2. Interactive controls:
   - Rotate: Click and drag
   - Zoom: Scroll or pinch
   - Pan: Right-click and drag
   - Reset view: Double-click
3. Download static PNG using camera icon in toolbar
4. HTML file is self-contained and shareable!
============================================================
```

### 5.2 Summary and outlook

This notebook started from the same ingredient as notebook *03_enzyme_oscillations* — a Hill-type enzymatic rate law with forward (substrate) inhibition — but followed it down a different path:

- Embedding the nonlinear rate law in an open-flow system (constant influx, linear removal) and reducing to a single ODE for $S$ gave an equation whose steady states could be found and classified analytically/numerically.
- For a range of the input parameter $a_1$ (roughly 0.51 to 1.15 in our numerical example), this equation has **three** steady states: two stable and one unstable — i.e. the system is **bistable**.
- Rewriting the equation as $dS/dt = -dV/dS$ turned this steady-state structure into a **potential landscape** $V(S)$: stable states are valleys, the unstable state is the barrier between them.
- Letting $a_1$ vary continuously turned the 1D landscape into a 2D surface $V(S, a_1)$ that visually reproduces **Waddington's epigenetic landscape**: a single valley splitting into two as a control parameter crosses a bifurcation point.

**Outlook.** Because this notebook's reduced model is a *single* ODE, the only possible long-term behaviours are steady states (stable or unstable) — oscillations are impossible in one dimension. Notebook *03_enzyme_oscillations* reintroduces the second variable (the product $P$) and replaces forward inhibition with *delayed, feedback* inhibition. With two coupled variables, a steady state can lose stability through a different route (a Hopf bifurcation), turning into an unstable spiral surrounded by a stable limit cycle — sustained oscillations. Multistability and oscillations are therefore best thought of as two outcomes of the same underlying nonlinear dynamics, reached by slightly different routes.

### 5.3 References and further reading


- Waddington, C.H. (1957). *The Strategy of the Genes*. George Allen & Unwin. — the original source of the "epigenetic landscape" metaphor used throughout this notebook.
- Ferrell, J.E. (2012). Bistability, bifurcations, and Waddington's epigenetic landscape. *Current Biology*, 22(11), R458–R466. — a modern, quantitative treatment connecting bistability and bifurcation theory directly to Waddington's picture.
- Gardner, T.S., Cantor, C.R., Collins, J.J. (2000). Construction of a genetic toggle switch in *Escherichia coli*. *Nature*, 403, 339–342. — an experimental realisation of a bistable gene-regulatory switch, the genetic analogue of the two-valley landscape above.
- Tyson, J.J., Chen, K.C., Novak, B. (2003). Sniffers, buzzers, toggles and blinkers: dynamics of regulatory and signaling pathways in the cell. *Current Opinion in Cell Biology*, 15(2), 221–231. — reviews recurring dynamical "modules" in cell regulation, includ-

ing toggle switches (bistability, this notebook) and blinkers/oscillators (notebook *03_enzyme_oscillations*). Available as a PDF in this folder.

- Wang, J., Zhang, K., Xu, L., Wang, E. (2011). Quantifying the Waddington landscape and biological paths for development and differentiation. *Proceedings of the National Academy of Sciences*, 108(20), 8257–8262. — a quantitative landscape approach applied to stem-cell differentiation, in the same spirit as the 2D landscape built in this notebook.